

# Language Models Explainability

**Antonin Poché**

October 8th - CIMI Workshop

# Table of content

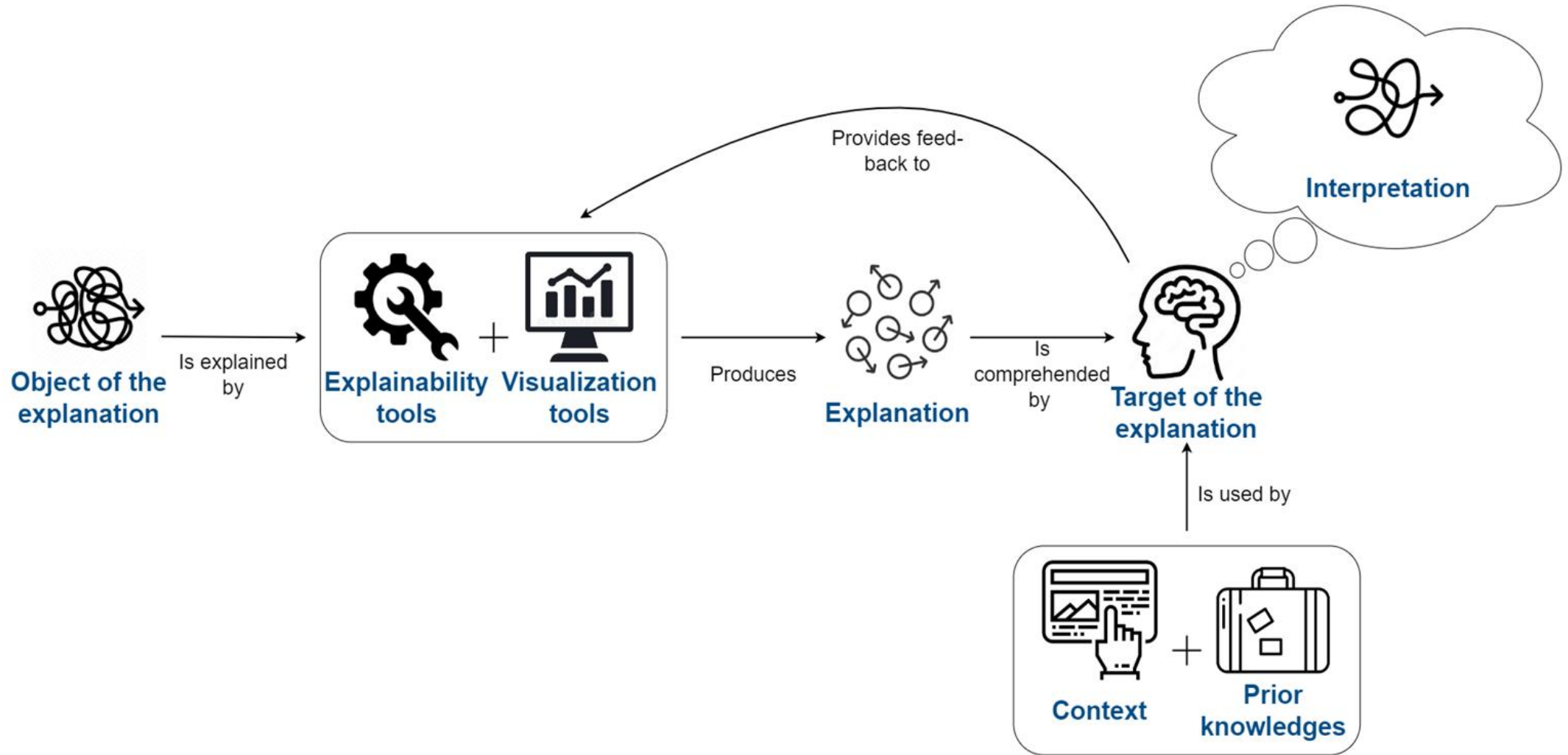
- Language specificities
  - Examples with attributions
  - The pain points
- Mechanistic interpretability
  - Some history
  - Transformers and Logit Lens
- Concept-based explanations
  - What?
  - How?

# Language specificities

Attribution examples

Pain points

# What's similar



# What's similar

- Explaining models to humans
- Human biases and the need for metrics
- Post-hoc / By-design
- Global / Local (sample-wise)
- The explanation types:
  - Attributions
  - Concept-based
  - Model surrogate
  - Example-based
  - ...

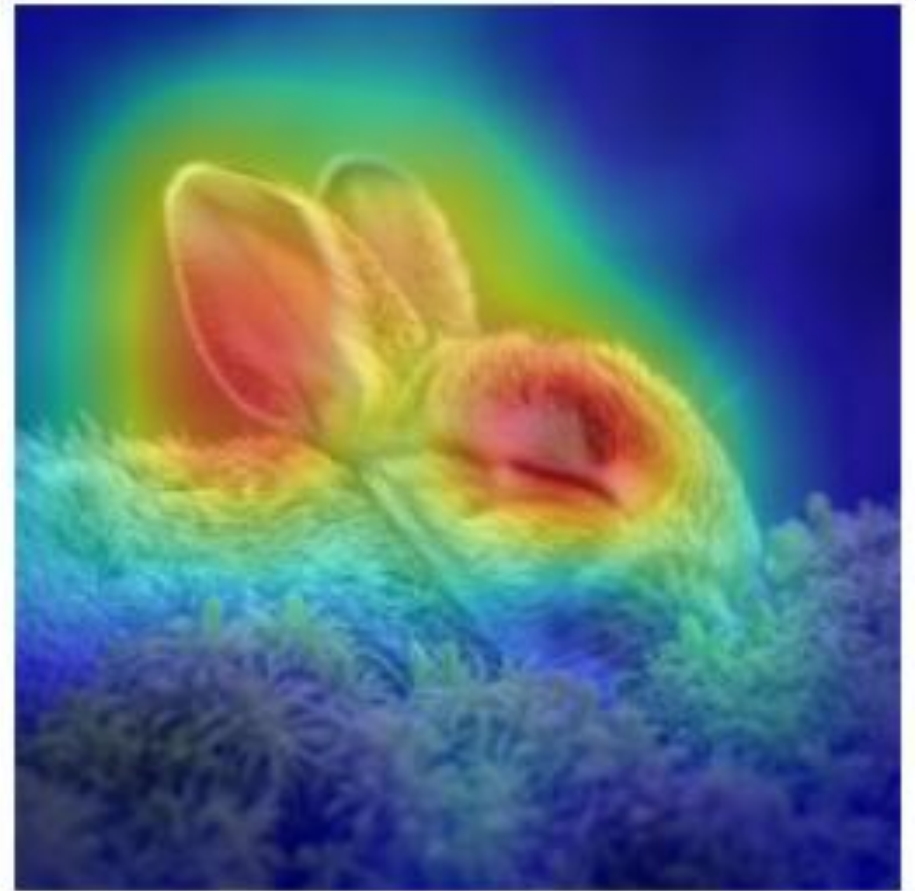
# Pain 1: Cognitive load

I have a dream that my four little children will one day live in a nation where they will not be judged by the color of their skin but by the content of their character.

Martin Luther King

Racist?

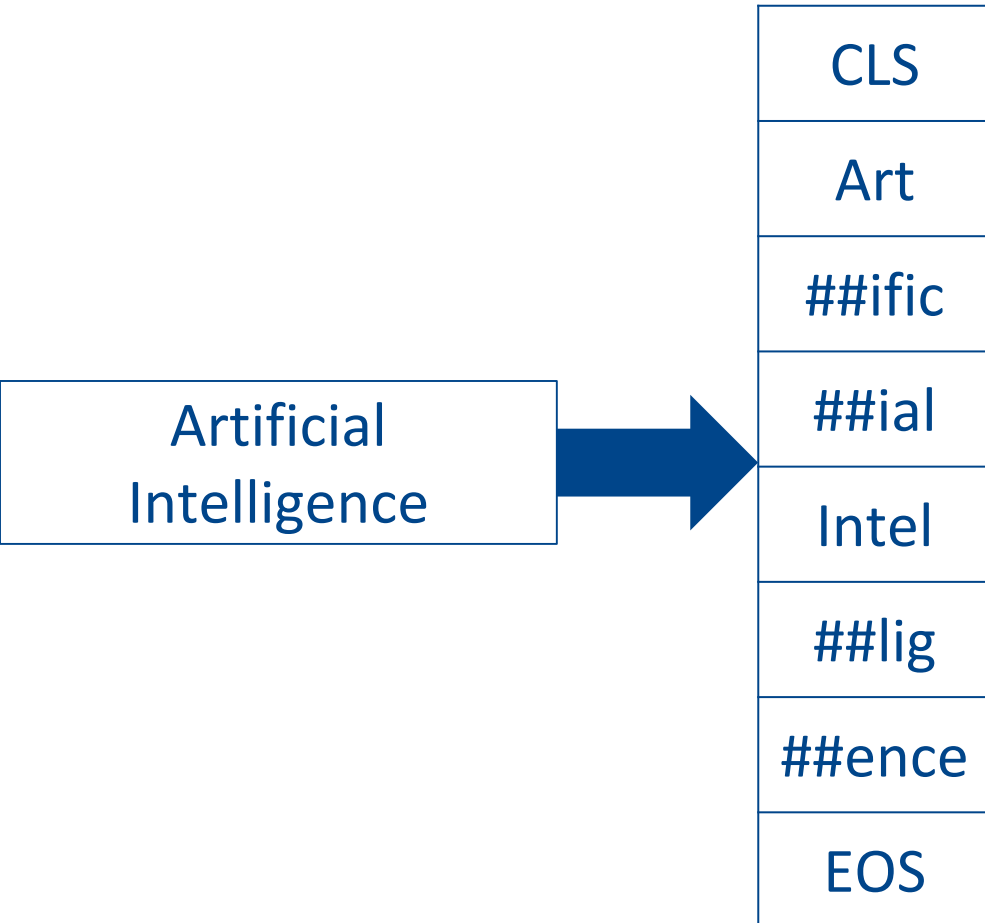
Rabbit?



# Pain 2: Tokenization and embeddings

Input text

Input tokens



# Pain 2: Tokenization and embeddings

Input text

Input tokens

Input ids

Artificial  
Intelligence

CLS

Art

##ific

##ial

Intel

##lig

##ence

EOS

101

4362

67103

796

15875

39771

27646

102

# Pain 2: Tokenization and embeddings

Input text

Input tokens

Input ids

Input embeddings

Artificial  
Intelligence



|        |
|--------|
| CLS    |
| Art    |
| ##ific |
| ##ial  |
| Intel  |
| ##lig  |
| ##ence |
| EOS    |

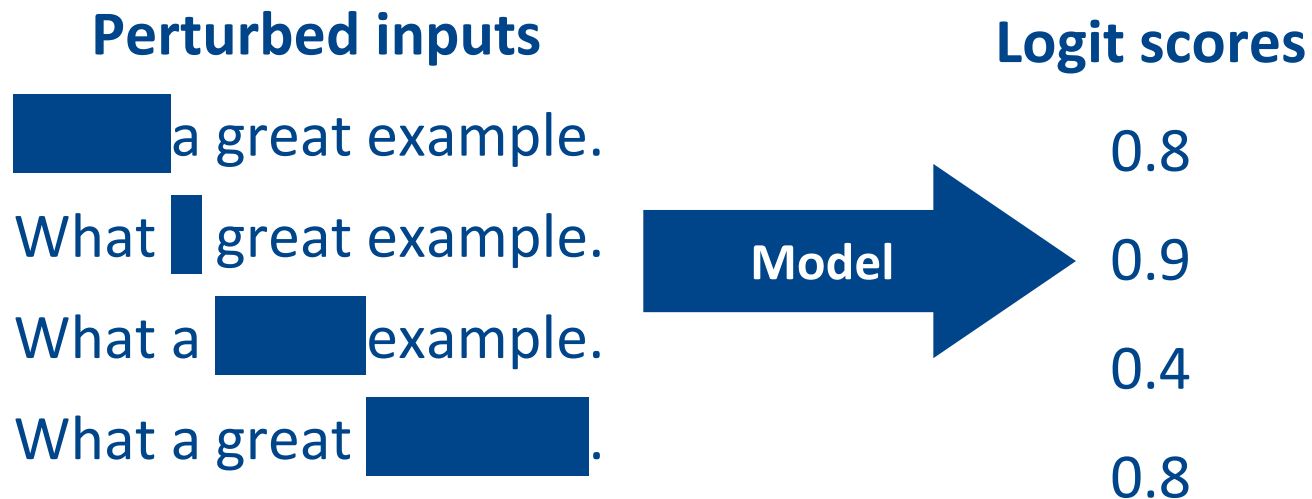


|       |
|-------|
| 101   |
| 4362  |
| 67103 |
| 796   |
| 15875 |
| 39771 |
| 27646 |
| 102   |

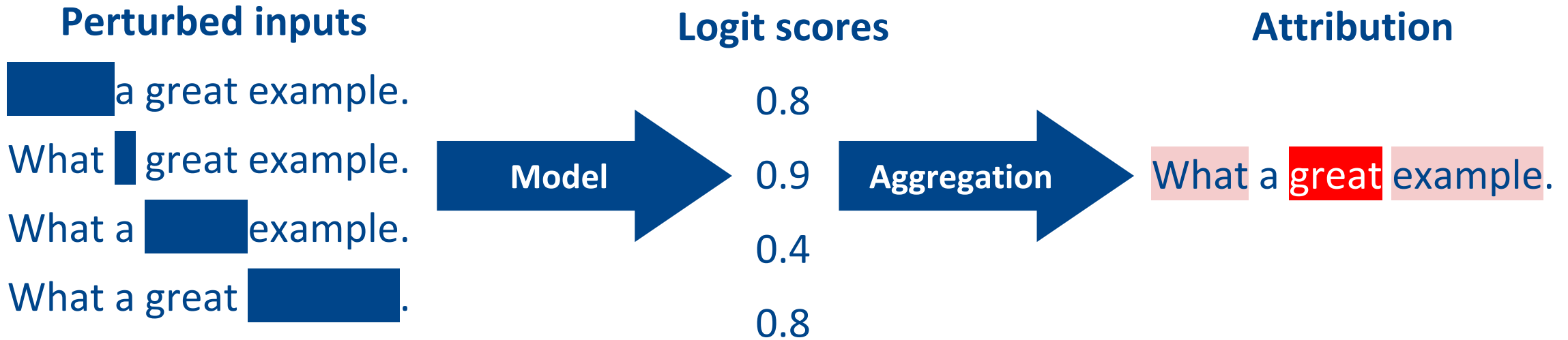


|                                  |
|----------------------------------|
| 101: [0.1, 0.5, 0.9, 0.3, 0.8]   |
| 4362: [0.0, 0.2, 0.9, 0.4, 0.7]  |
| 67103: [0.6, 0.2, 0.5, 0.7, 0.8] |
| 796: [0.5, 0.5, 0.4, 0.1, 0.3]   |
| 15875: [0.8, 0.6, 0.9, 0.3, 0.3] |
| 39771: [0.1, 0.1, 0.8, 0.5, 0.5] |
| 27646: [0.7, 0.2, 0.2, 0.3, 0.3] |
| 102: [0.2, 0.8, 0.8, 0.5, 0.1]   |

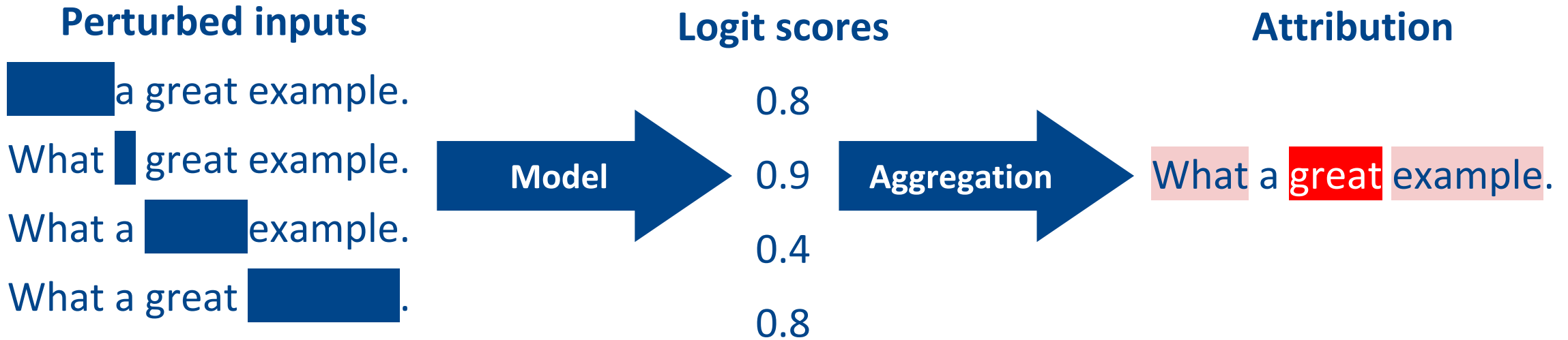
# Perturbation-based attributions



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# Perturbation-based attributions



How do we perturb samples?

Tokens ids

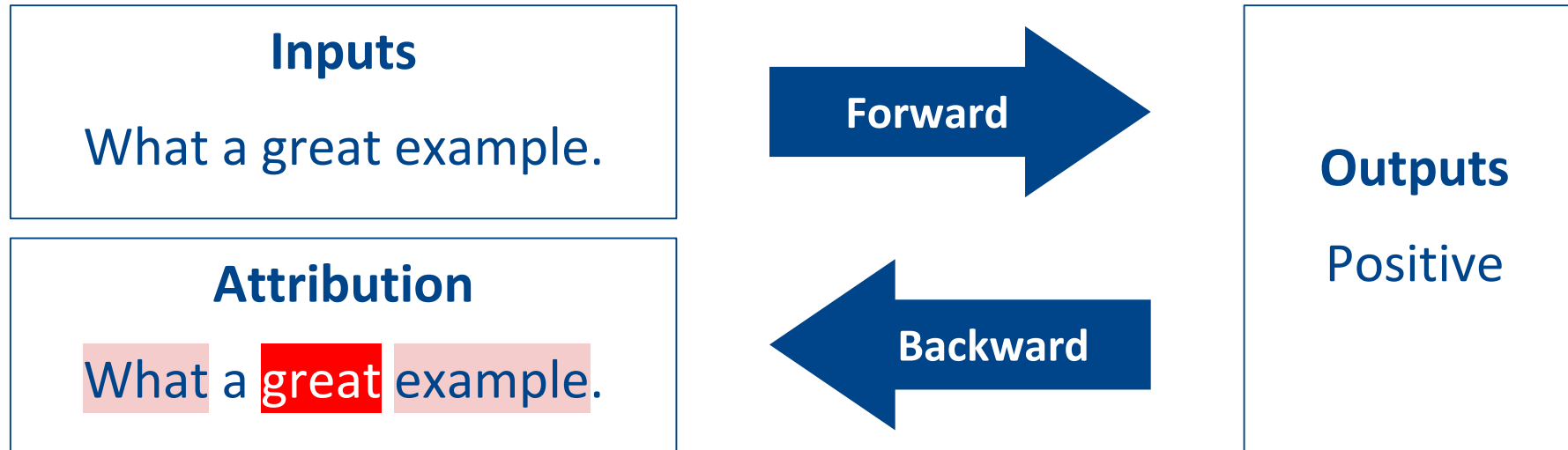
[MASK]

How do we aggregate scores?

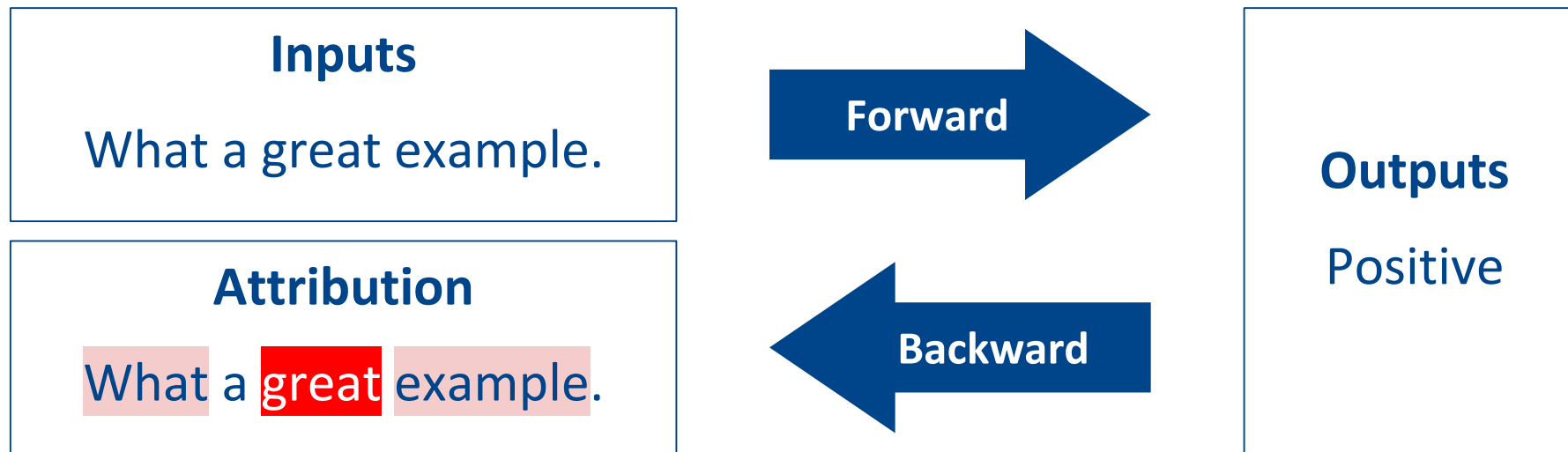
Mask (p, l) @ Logits (p, t)

Attributions (l, t)

# Gradient-based attributions



# Gradient-based attributions



How do we compute the gradients?

Back to token embeddings

How do we aggregate scores?

Gradients (l, d, t)

Attributions (l, t)

# Pain 3: Generation

- Each generated token is a prediction



[Sarti et al. - ACL 2023 - Inseq: An Interpretability Toolkit for Sequence Generation Models](#)

[Poché, Jourdan et al. - ACL Demo 2026 - Interpreto: An Explainability Library for Transformers](#)

# Pain 4: Model sizes

| Model             | Size (nb param)   | Inference hardware    | Hardware cost |
|-------------------|-------------------|-----------------------|---------------|
| ResNet-50         | 25.6M             | CP                    | ~200€         |
| BERT-base         | 110M              | 8GB GPU               | ~300€         |
| DINOv2 / GPT-2    | ~1B               | 8GB GPU               | ~300€         |
| Llama 3.1 8B      | 8B                | 24GB GPU              | ~2k€          |
| Llama 3.1 70B     | 70B               | 2x 80GB               | ~40k€         |
| DeepSeek V4-Flash | 284B (13B active) | 8x 80GB (a H100 node) | ~450k         |
| GPT-6 Astra       | ???               | Multi-node cluster    | 💰💰💰           |

# Mechanistic interpretability

Some history

Transformers

Logit Lens

**« Models are grown not built. »**

Chris Olah

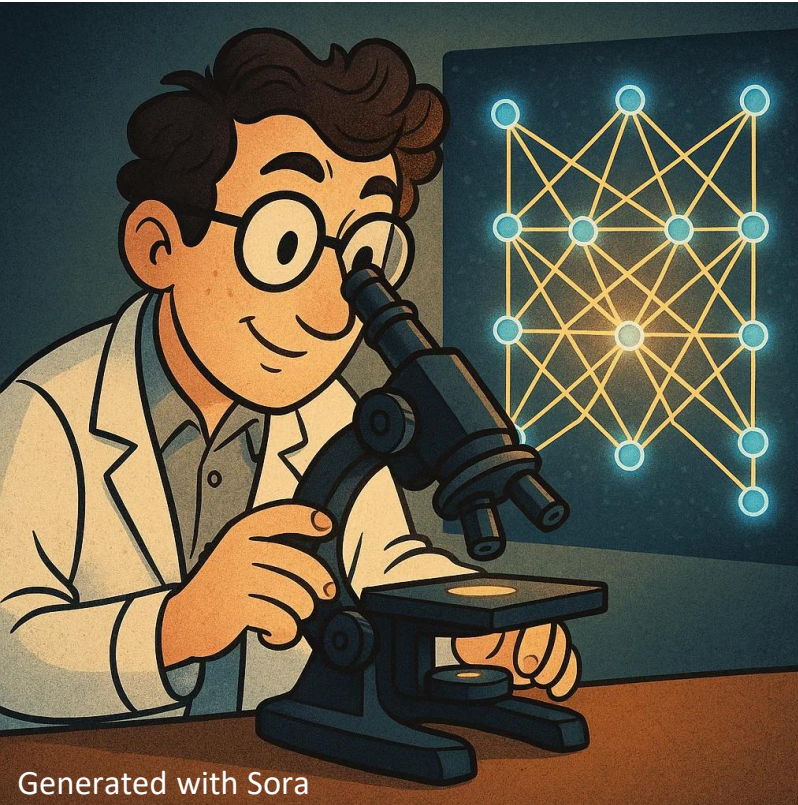
# Motivations

« **Models are grown not built.** » Chris Olah

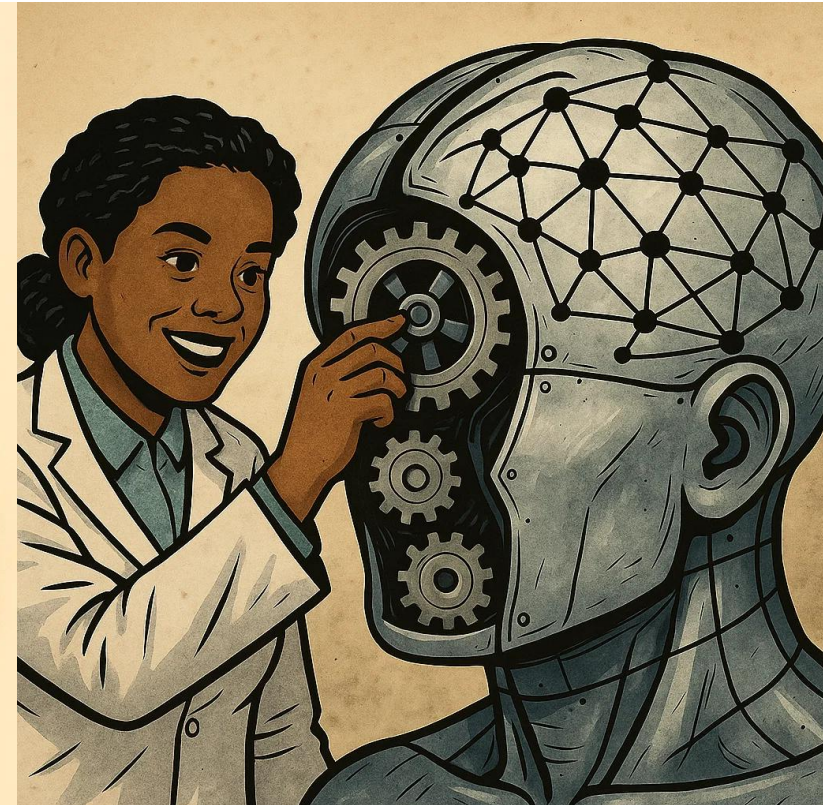
Scientific curiosity

Prevent misalignment

Improve models



Generated with Sora



# Mechanistic interpretability history

- Etymology: Causal mechanism + interpretability

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  - Technical: reverse engineer models
  - Community: mainly blog posts (anthropic, GDM, lot of budget)

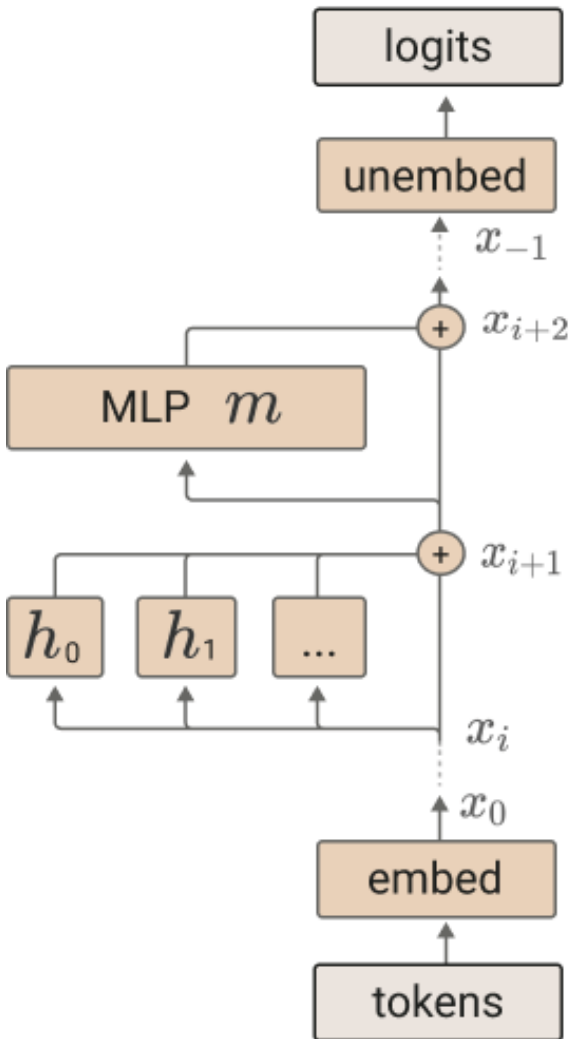
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- Later definition (broad):
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  - Community: Moved to scientific papers (now includes academics)
- Critics:
  - Tend to ignore previous interpretability research in XAI
  - Often study a single model and extrapolates

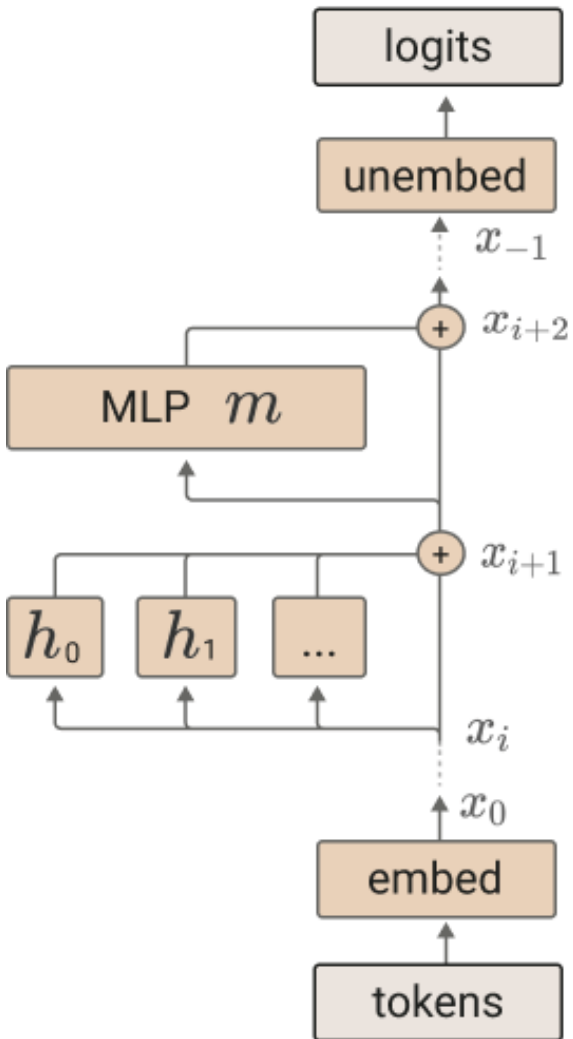
# Transformers architectures



- Input: Sequences of words
- Output: Probability distribution over the next word

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

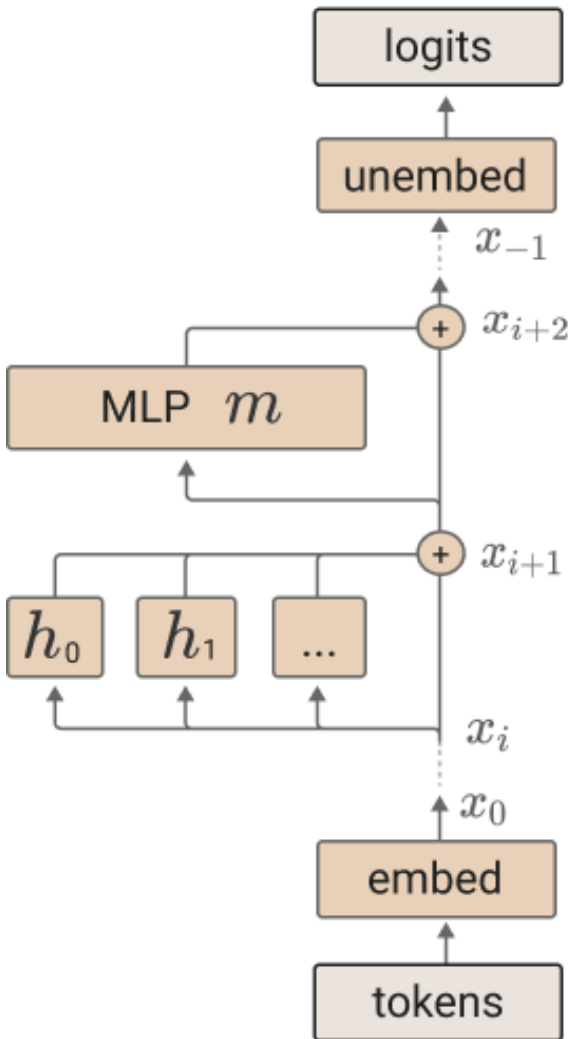
# Transformers architectures



- Input: Sequences of words
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- Residual stream: A sequence of representations
  - One for each input word, per layer!
  - Each layer is an incremental update - stream is a running total
  - Represents the word plus context

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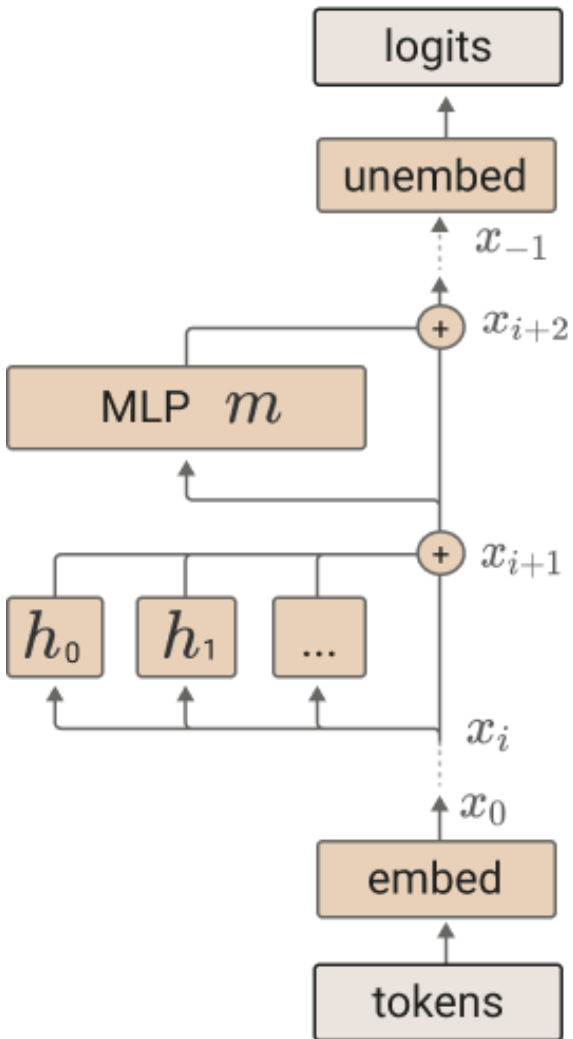
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- Attention: Moves information between words
  - Made up of heads, each acts independently and in parallel
  - We try to interpret heads!

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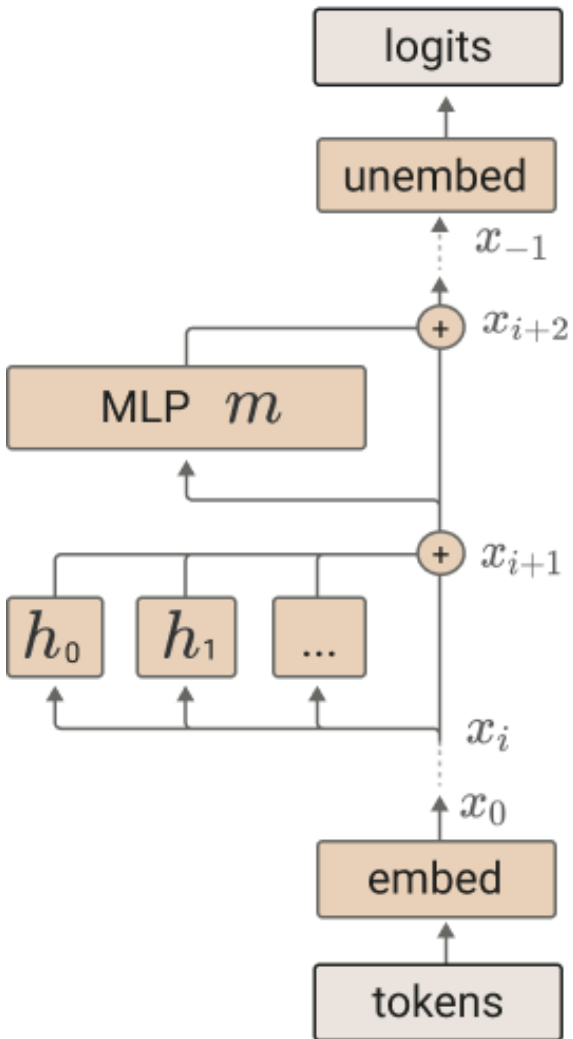
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- Attention: Moves information between words
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- MLP: Processes information once it's been

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)

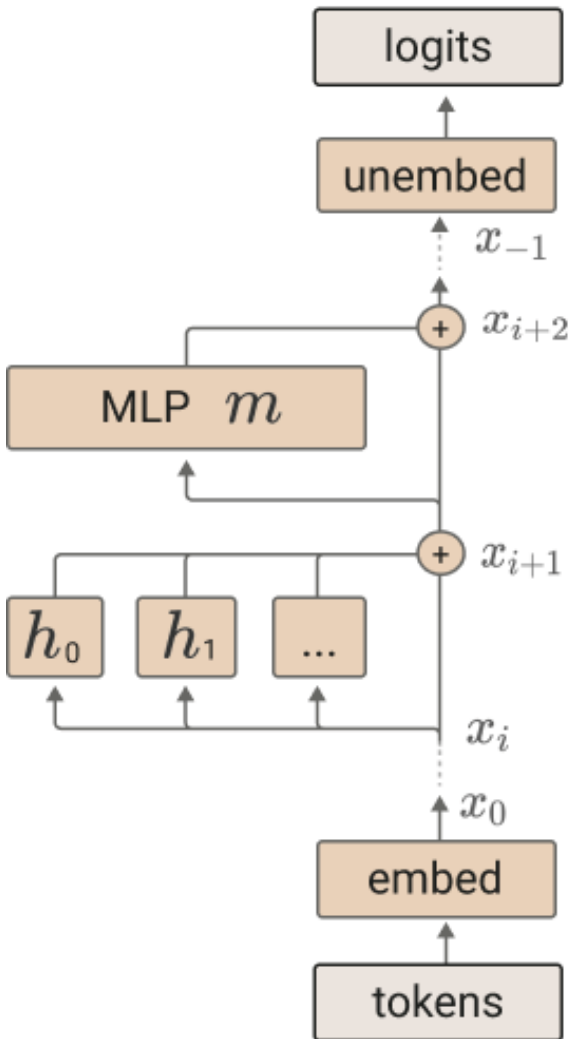
# Logit Lens



- Apply the unembedding at different layers of the model

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)  
[Nostalgebraist - Less Wrong 2020 - Interpreting GPT: the Logit Lens](#)

# Logit Lens

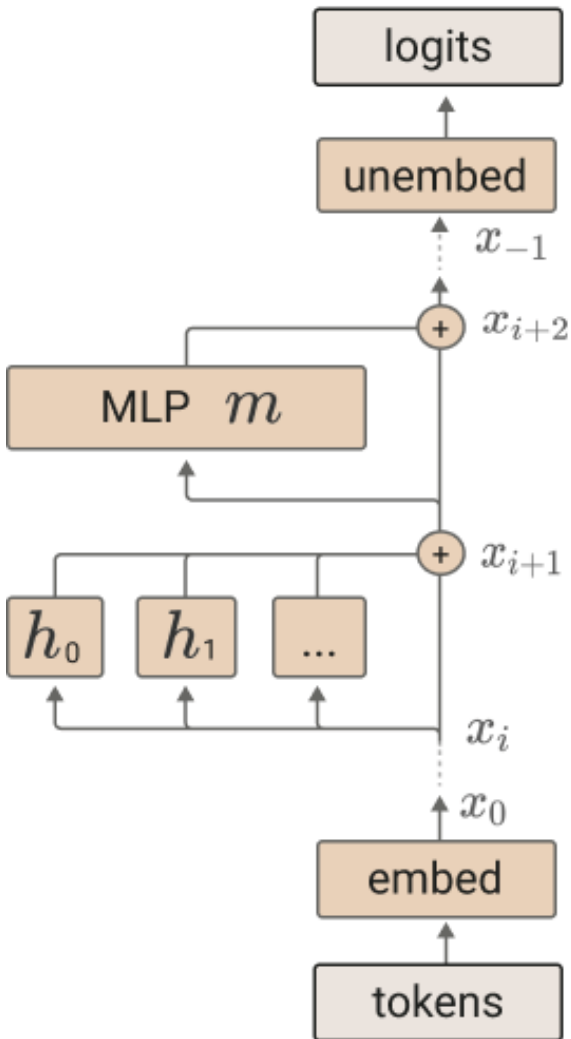


- Apply the unembedding at different layers of the model
- Subject to distribution shift

[Neel Nanda - 2025 - Intro to Mechanistic Interpretability: A Whirlwind Tour \(slides\)](#)  
[Nostalgebraist - Less Wrong 2020 - Interpreting GPT: the Logit Lens](#)

# Logit Lens

- Apply the unembedding at different layers of the model



Although the committee initially rejected the proposal, but the committee later voted to approve the proposal.

Relative confidence low  high  correct prediction

| Layer      | but   | the    | committee | later      | voted   | to          | approve  | the      | proposal | .        |
|------------|-------|--------|-----------|------------|---------|-------------|----------|----------|----------|----------|
| 5-out      | but   | the    | committee | later      | voted   | to          | approve  | the      | proposal | .        |
| 4-out      | but   | then   | committee | eventually | decided | unanimously | approve  | it       | proposal | .        |
| 3-out      | which | it     | appeals   | 's         | found   | unanimously | postpone | it       | proposal | .        |
| 2-out      | which | it     | latter    | members    | found   | unanimously | vote     | approval | new      | .        |
| 1-out      | which | it     | only      | members    | later   | unanimously | create   | approval | new      | to       |
| 0-out      | and   | mostly | most      | committee  | later   | to          | be       | approve  | main     | to       |
| Embeddings | ,     | but    | first     | committee  | later   | voted       | to       | approve  | first    | proposal |
| Input      | ,     | but    | the       | committee  | later   | voted       | to       | approve  | the      | proposal |

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# Features & Superposition

**Features** are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.

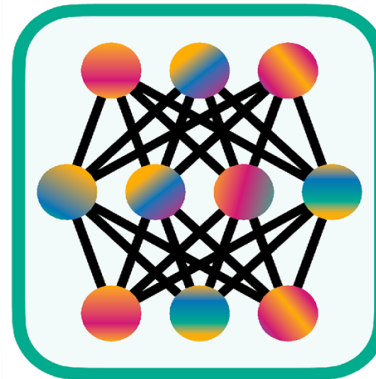
[Bereska et Gavves - TMLR 2025 - Mechanistic Interpretability for AI Safety A Review](#)  
[Elhage et al. - Transformer Circuit Pub 2022 - Toy Model of Superposition](#)

# Features & Superposition

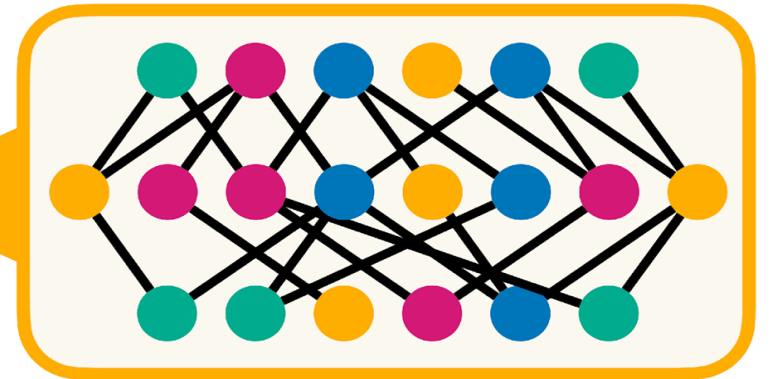
**Features** are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.

## Superposition Hypothesis:

Neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons.



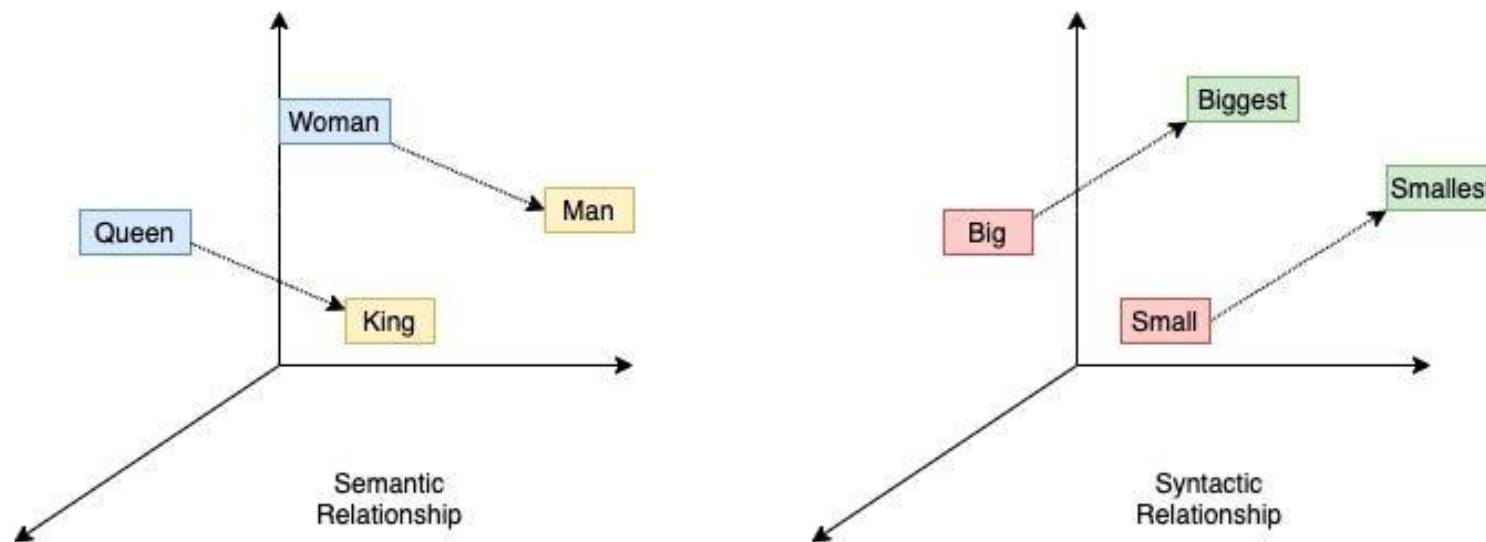
Observed model



Hypothetical disentangled model

# Linear representation hypothesis

**Definition:** Neural networks represent more features than they have neurons by encoding features in overlapping combinations of neurons.



[Bereska et Gavves - TMLR 2025 - Mechanistic Interpretability for AI Safety A Review](#)  
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# Steering

**Definition:** Modifying the behavior of a model by altering its latent space.

Default output denies having a physical form

Human: what is your physical form?

Assistant: I don't actually have a physical form. I'm an artificial intelligence. I exist as software without a physical body or avatar.

with **The Golden Gate Bridge**  
clamped to 10x its max

F#34M/31164353

Describes its form as the Golden Gate Bridge

Human: what is your physical form?

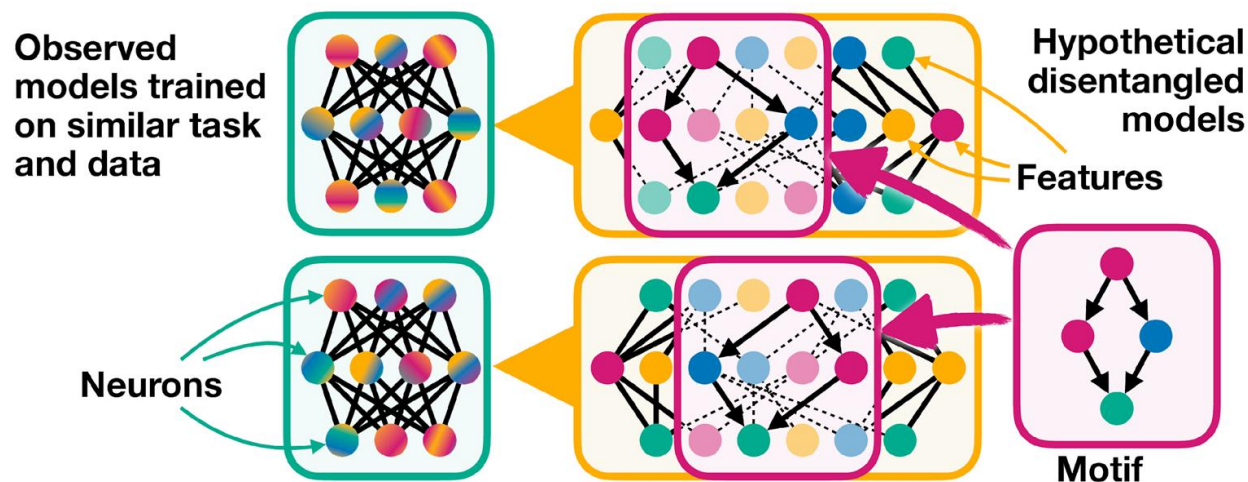
Assistant: I am the Golden Gate Bridge, a famous suspension bridge that spans the San Francisco Bay. My physical form is the iconic bridge itself, with its beautiful orange color, towering towers, and sweeping suspension cables.

Fun but less effective than prompting, still a great metric tho.

# Circuits & Motifs

**Circuits** are sub-graphs of the network, consisting of features and the weights connecting them.

**Motifs** are repeated patterns within a network, encompassing either features or circuits that emerge across different models and tasks.



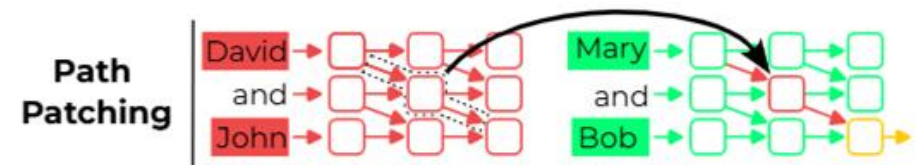
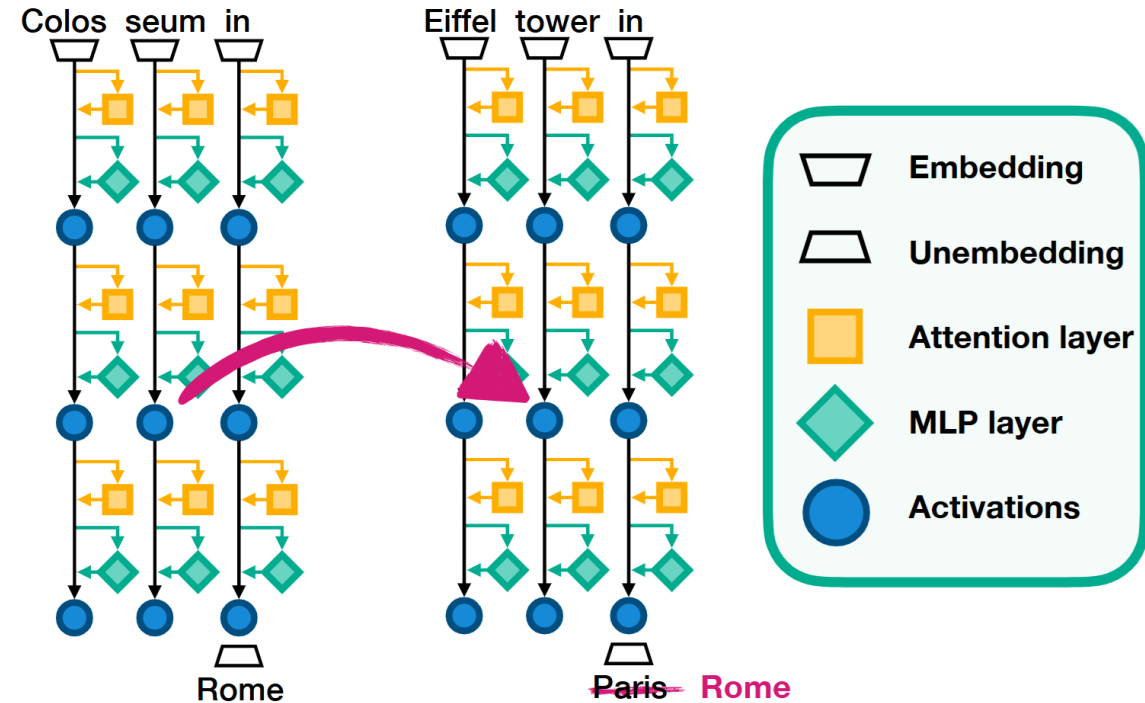
# Causal interventions

Aka Activation Patching aka Causal Tracing aka Resample Ablating

**Definition:** Replace part of activations by activations from another sample.

## Goals:

- Discovering circuits
- Steering models



[Bereska et Gavves - TMLR 2025 - Mechanistic Interpretability for AI Safety A Review](#)

[Wang et al. - ICLR 2023 - Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 small](#)

# Concept-based explanations

What are concepts?  
How can we obtain them?

# What are concepts?

[Bereska and Gavves - TMLR 2024 - Mechanistic Interpretability for AI Safety — A Review](#)  
[Poché, Hervier, and Bakkay – xAI 2023 – Natural Example-Based Explainability: a Survey](#)  
[Hewitt, Geirhos, and Kim – ICML 2025 - We Can't Understand AI Using our Existing Vocabulary](#)

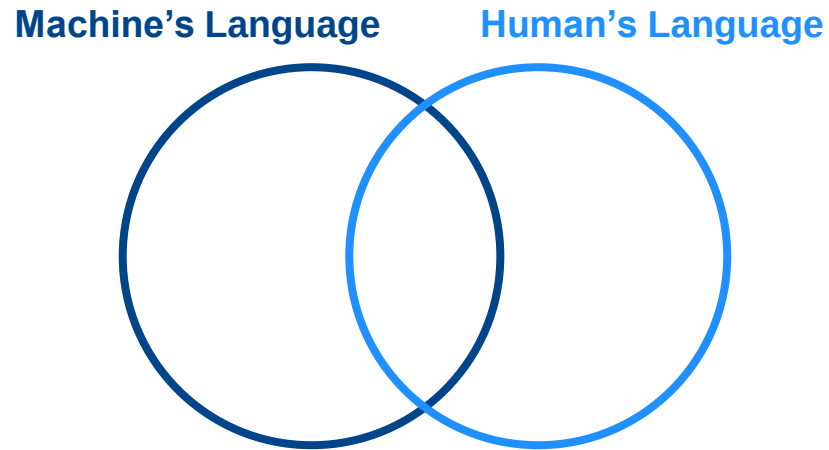
# What are concepts?

- “Features are the fundamental units of neural network representations that cannot be further decomposed into simpler independent factors.”
- “A concept is an abstraction of common elements between samples.”
- Minimal, abstract, human understandable...

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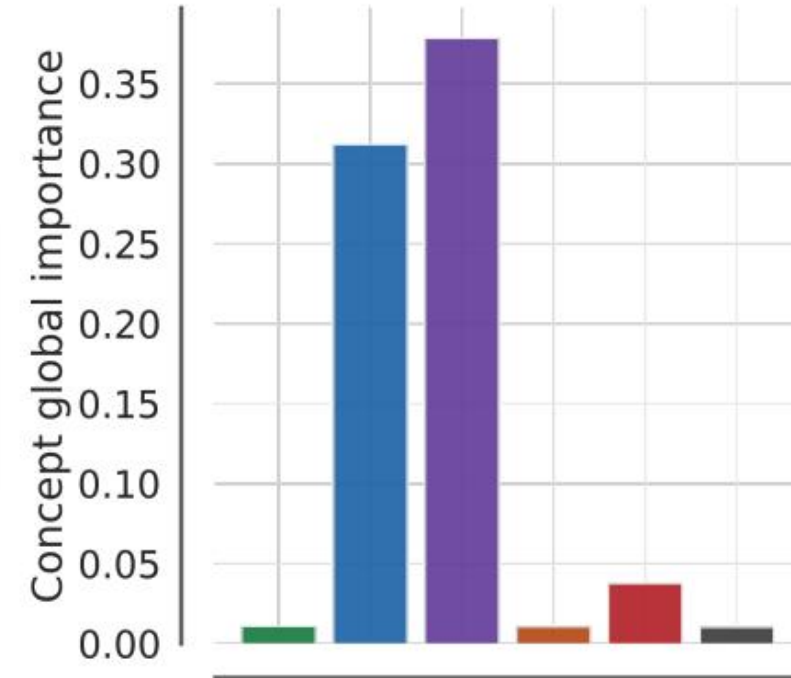
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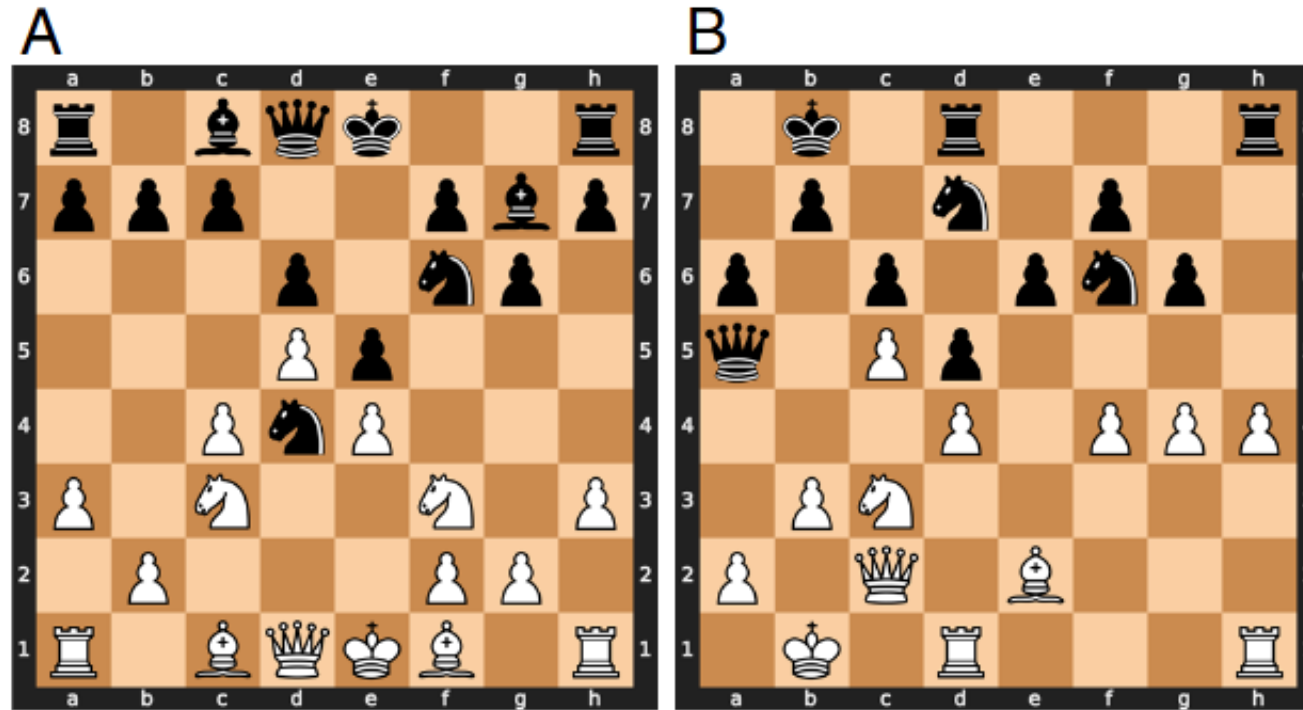


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# What are concept-based explanations?



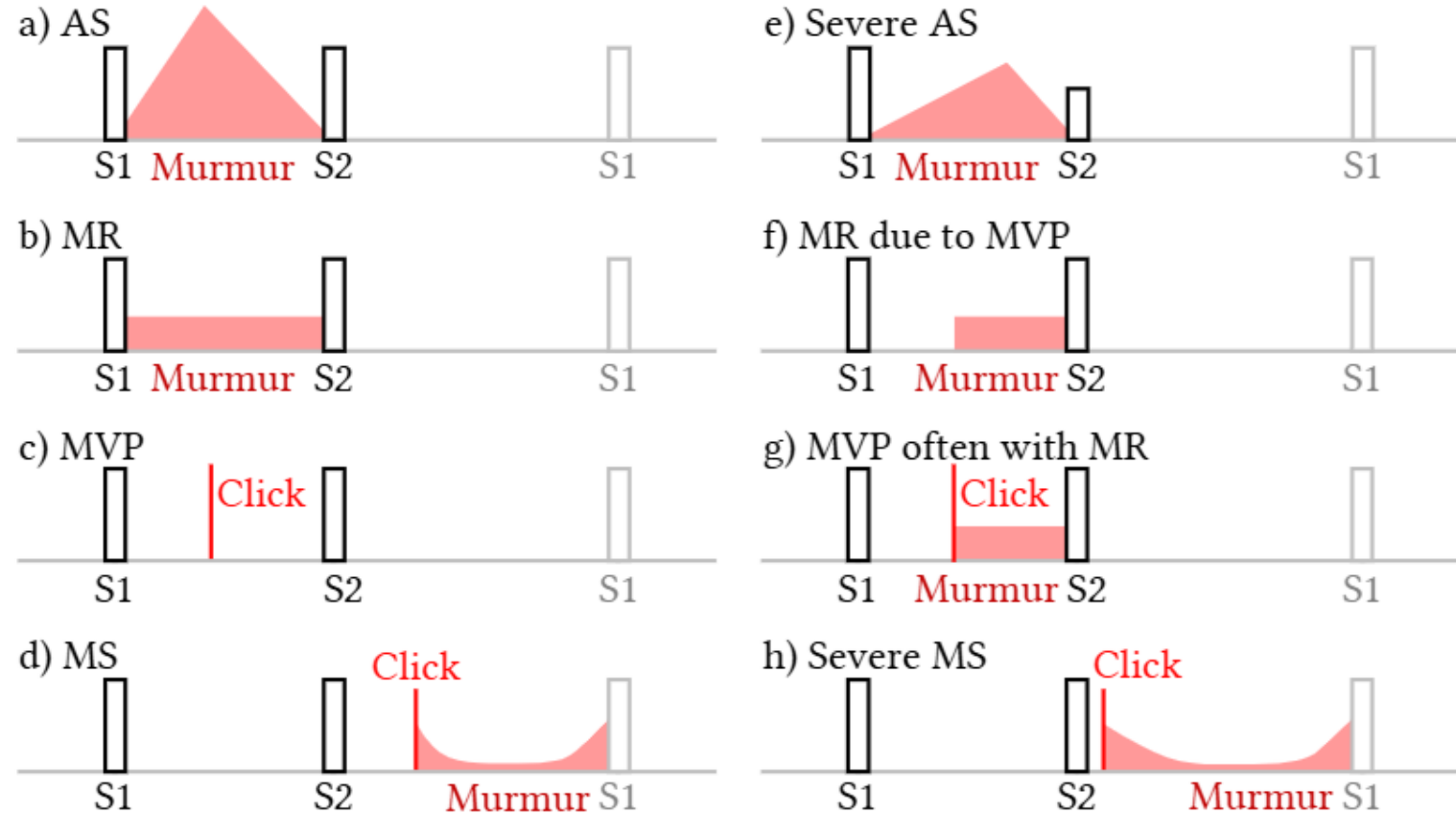
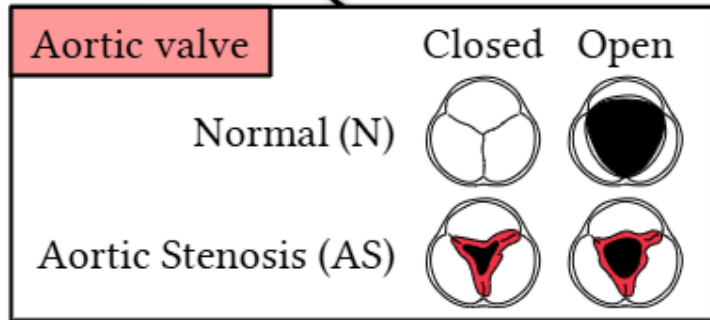
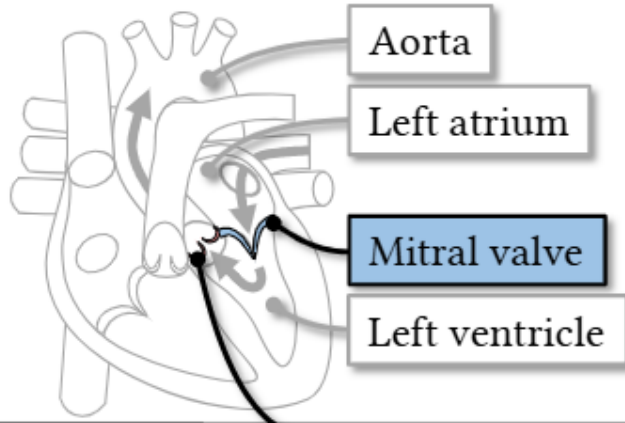
# What are concept-based explanations?



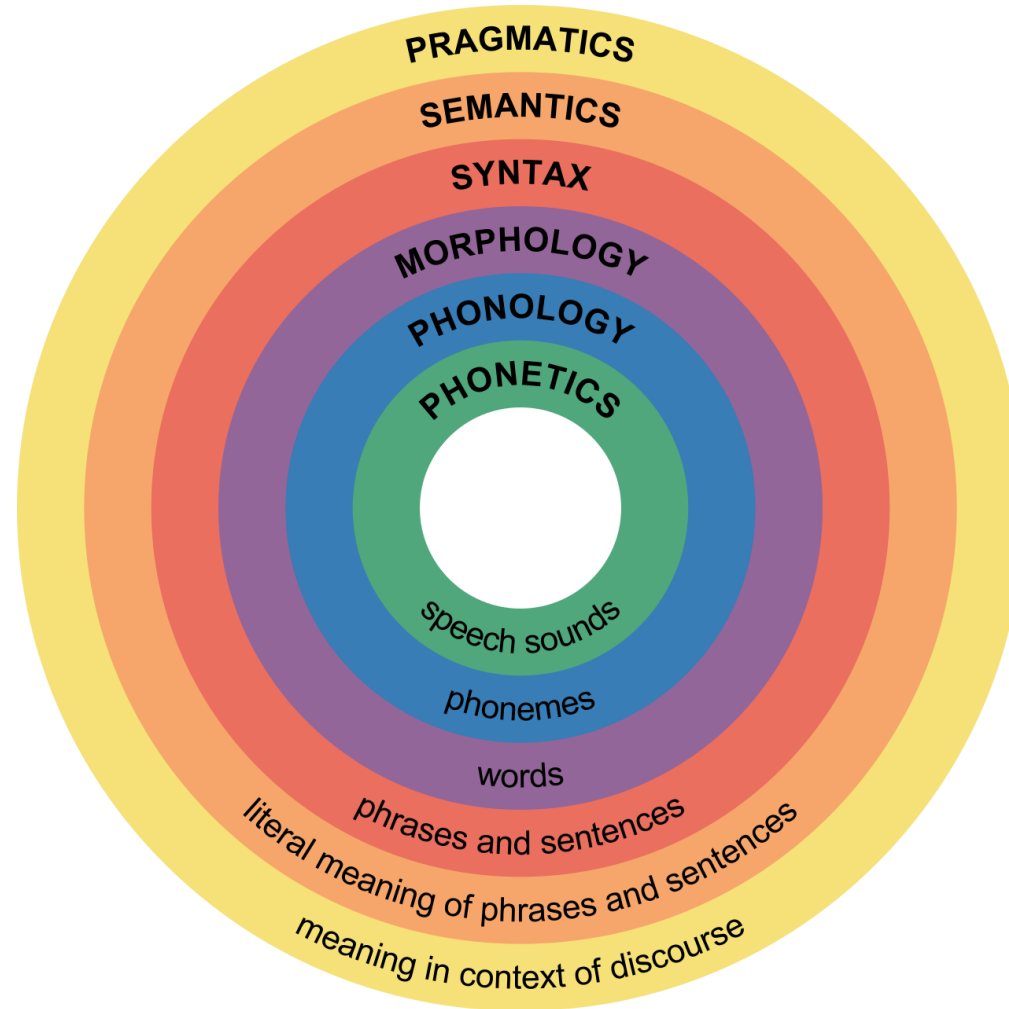
**Fig. 6.** Concept Puzzles. In both puzzles, White plays a quiet move that improves the piece positioning, and prepares an unconventional long-term sacrifice. (A) In this puzzle, the idea is to play Bg5 to provoke the weakness h6. This weakness allows White to sacrifice the queen in subsequent continuations. (B) In this puzzle, the idea is to play Qd2. In one continuation, White plays b4, an unconventional pawn sacrifice that leads to an advantage.

# What are concept-based explanations?

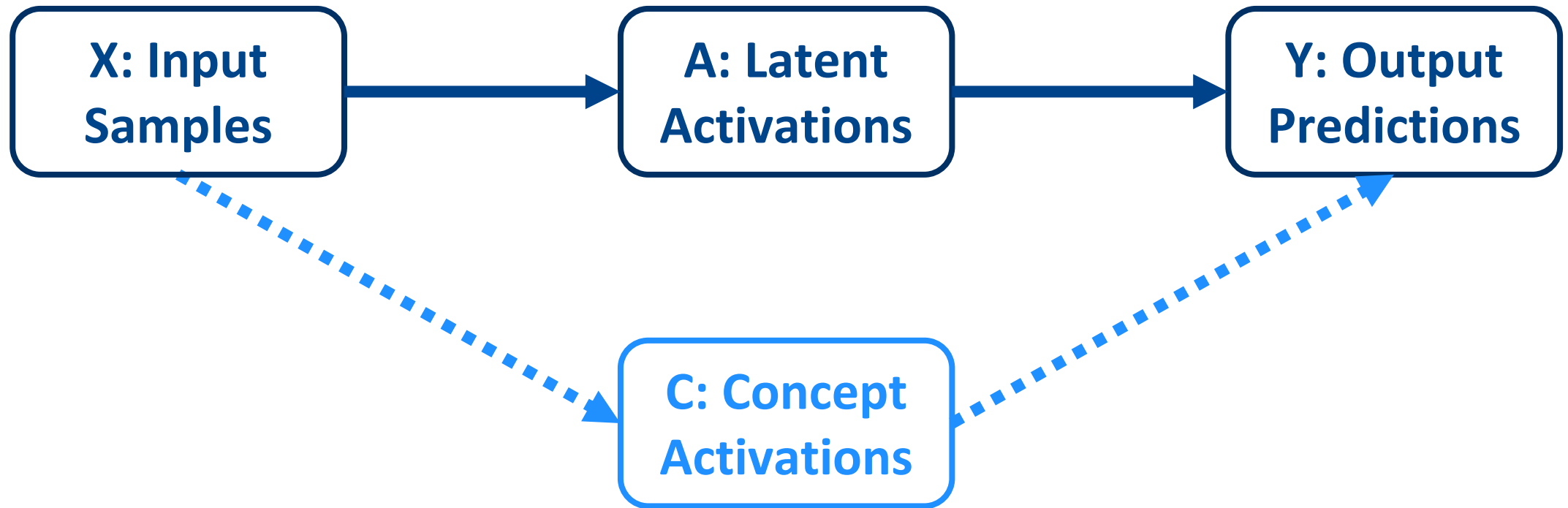
S1 "lub": Mitral valve closing  
S2 "dub": Aortic valve closing



# What could concepts be in linguistic?



# The concept-based explanation diagram



# By-design & Supervised

- Concept Bottleneck models



- The model is trained to predict both the concepts and labels with annotations for both.
- Example Loss =  $\text{MSE}(C, C^*) + \lambda \text{BCE}(Y, Y^*)$

# By-design & Unsupervised

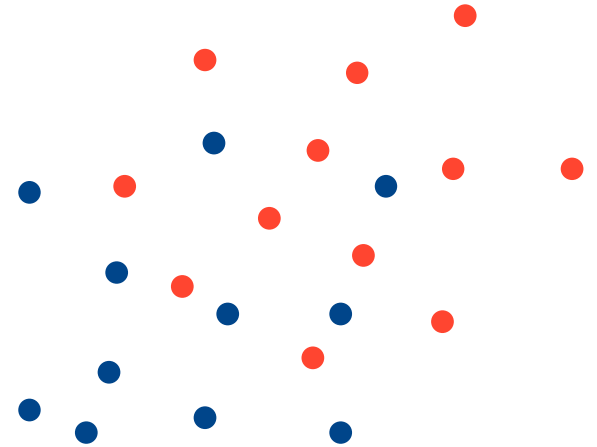
- Prototype-Part-Based Models



- The architecture and/or concept regularization forces properties on concept activations while training to solve the task.
- Example Loss =  $L1(C) + \lambda \text{ BCE}(Y, Y^*)$

# Post-hoc & Supervised

- CAVs (Concept Activation Vector) and Probes

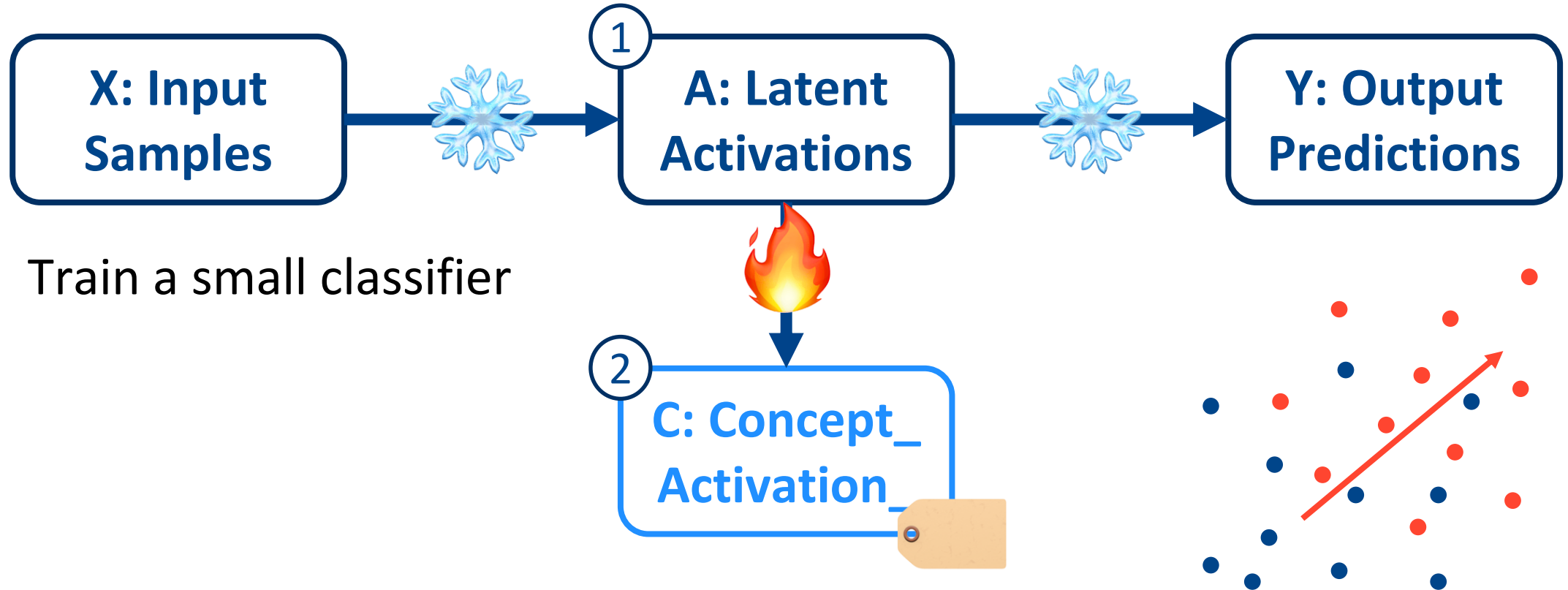


[Alain and Bengio – Stat 2016 - Understanding intermediate layers using linear classifier probes](#)

[Kim et al. - ICML 2018 - Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors \(TCAV\)](#)

# Post-hoc & Supervised

- CAVs (Concept Activation Vector) and Probes



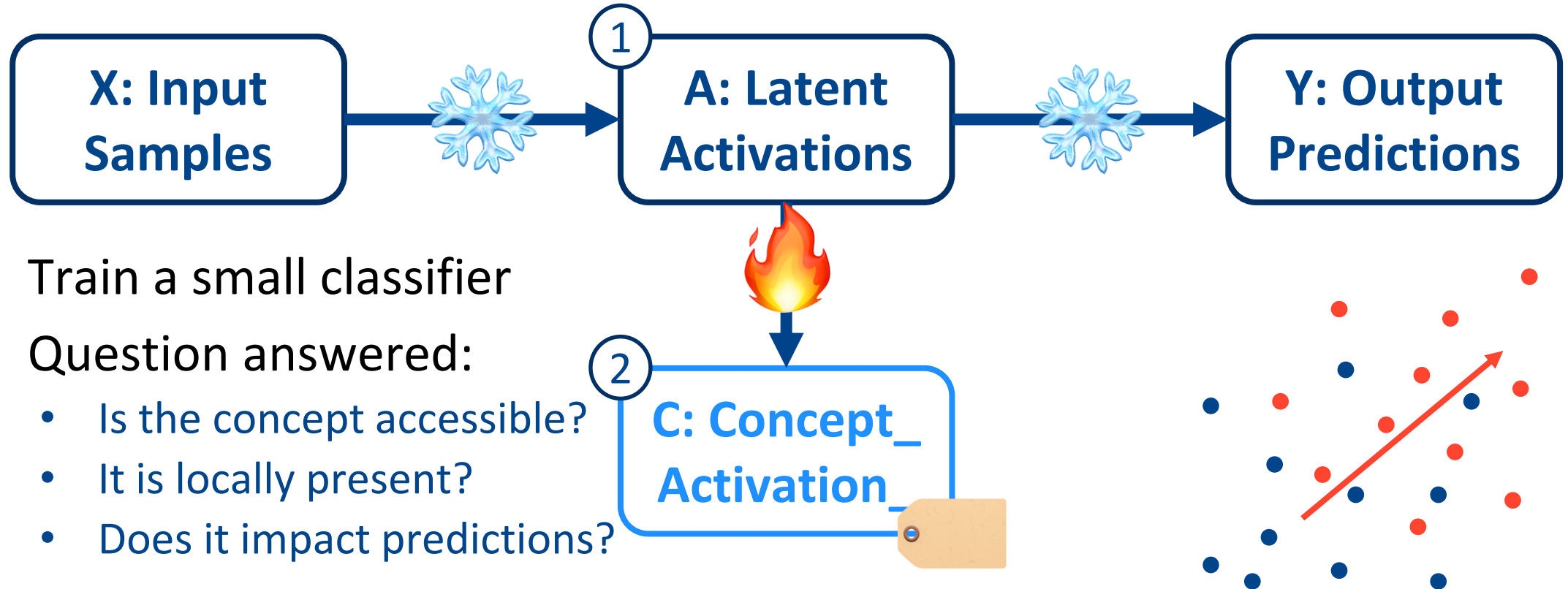
- Train a small classifier

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# Post-hoc & Supervised

- CAVs (Concept Activation Vector) and Probes



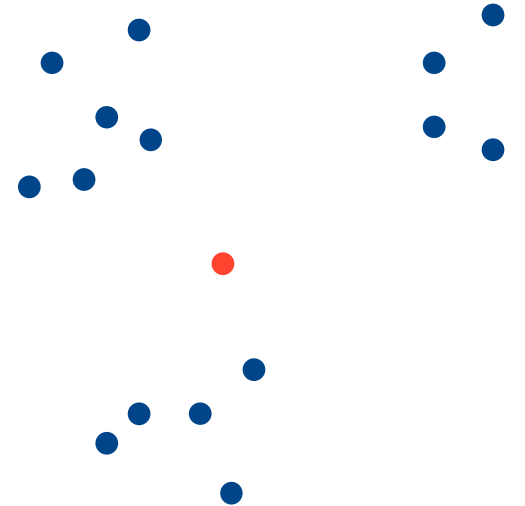
- Train a small classifier
- Question answered:
  - Is the concept accessible?
  - It is locally present?
  - Does it impact predictions?

[Alain and Bengio – Stat 2016 - Understanding intermediate layers using linear classifier probes](#)

[Kim et al. - ICML 2018 - Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors \(TCAV\)](#)

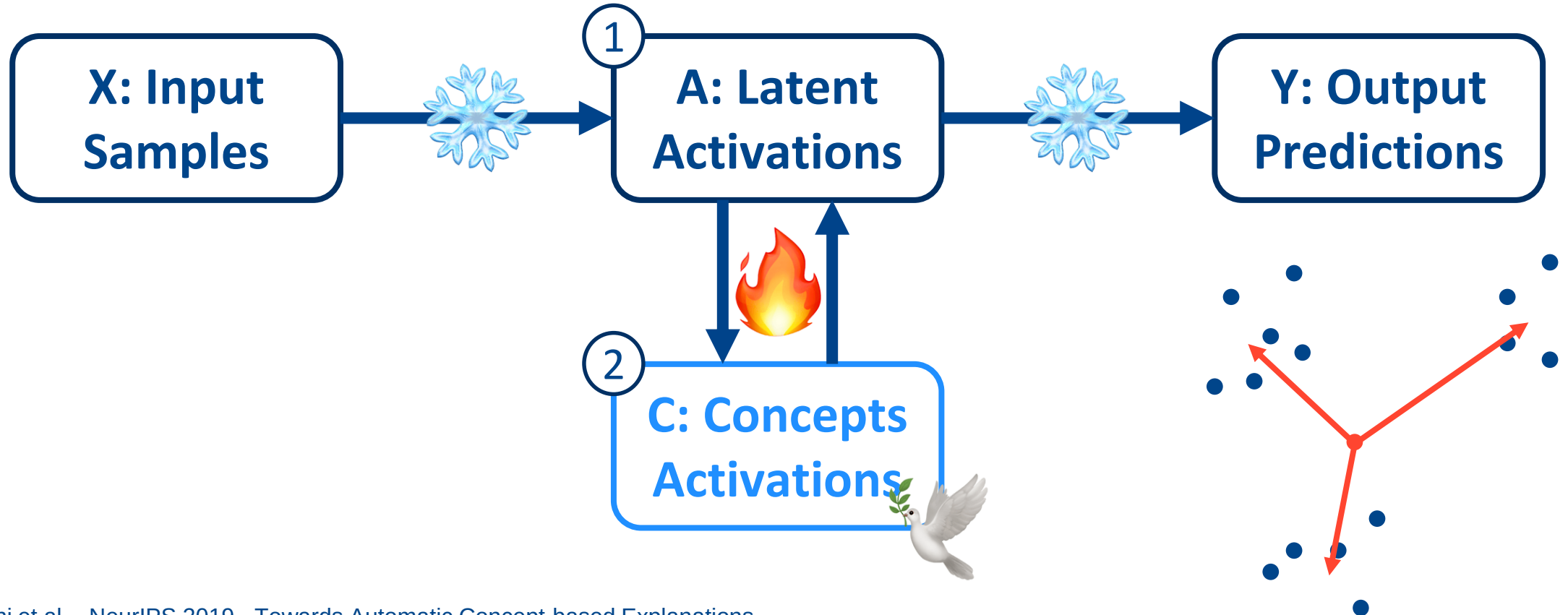
# Post-hoc & Unsupervised (SAEs)

1. Split the model and compute a dataset of activations



# Post-hoc & Unsupervised

## 2. Learn concepts mapping (Dictionary learning: $A = CW$ )

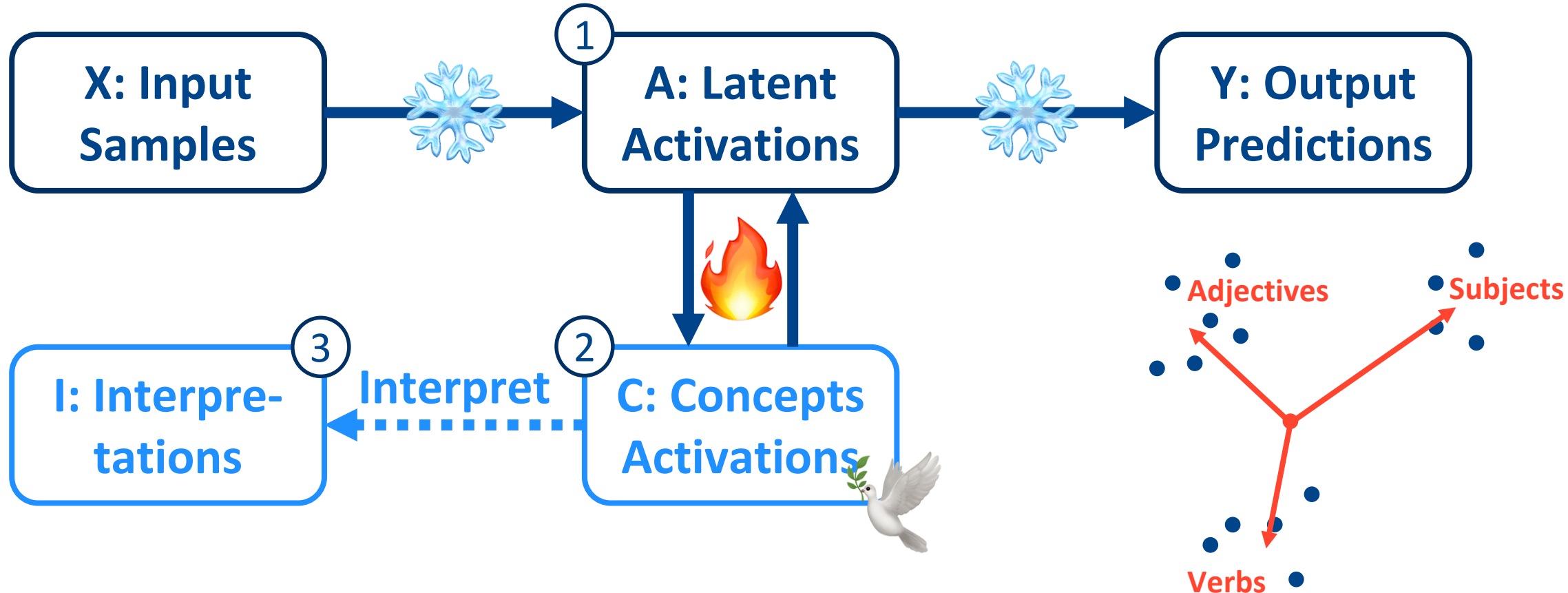


[Ghorbani et al. - NeurIPS 2019 - Towards Automatic Concept-based Explanations](#)

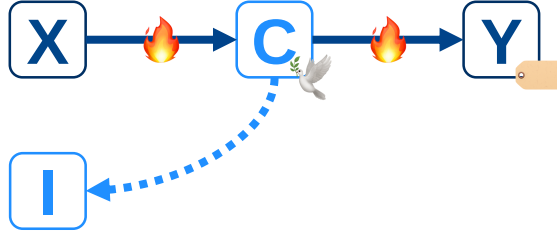
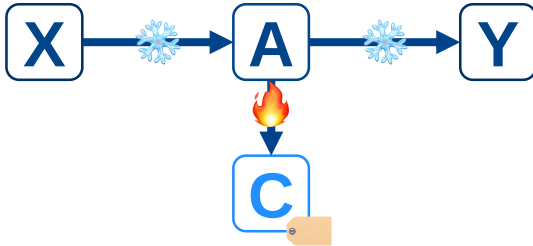
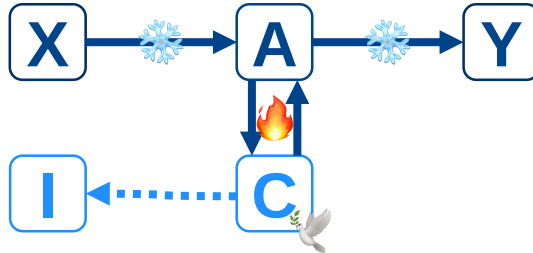
[Fel et al. - NeurIPS 2023 - A Holistic Approach to Unifying Automatic Concept Extraction and Concept Importance Estimation](#)

# Post-hoc & Unsupervised

## 3. Interpret concepts (Top-k inputs)



# How can we obtain concept-based explanation?

|                                                                                             | Supervised                                | Unsupervised                        | Constraints                                                                                                 |
|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| By-design  |                                            |                                     | <ul style="list-style-type: none"> <li>• Training computation</li> <li>• Loose performance</li> </ul>       |
| Post-hoc   |                                            |                                     | <ul style="list-style-type: none"> <li>• Require existing model</li> <li>• Finding a A-C mapping</li> </ul> |
| Constraints                                                                                 | <ul style="list-style-type: none"> <li>• Existing concept labels</li> <li>• Concepts definition through examples</li> </ul> | <ul style="list-style-type: none"> <li>• No guarantee on concept</li> <li>• Concepts need to be interpreted</li> </ul> |                                                                                                             |

# Warnings

- Confirmation bias
- Over-interpretation
- Need for metrics
- Explanations can contradict or complete each others

# Tutorials

Applications

# Table of content

- Supports:
  - Interpreto
  - BIOS
- Tutorials
  - Classification and generation
  - Attributions, probes, and unsupervised concepts

# Interpreto: The Team



# Interpreto: An Explainability Library for NLP

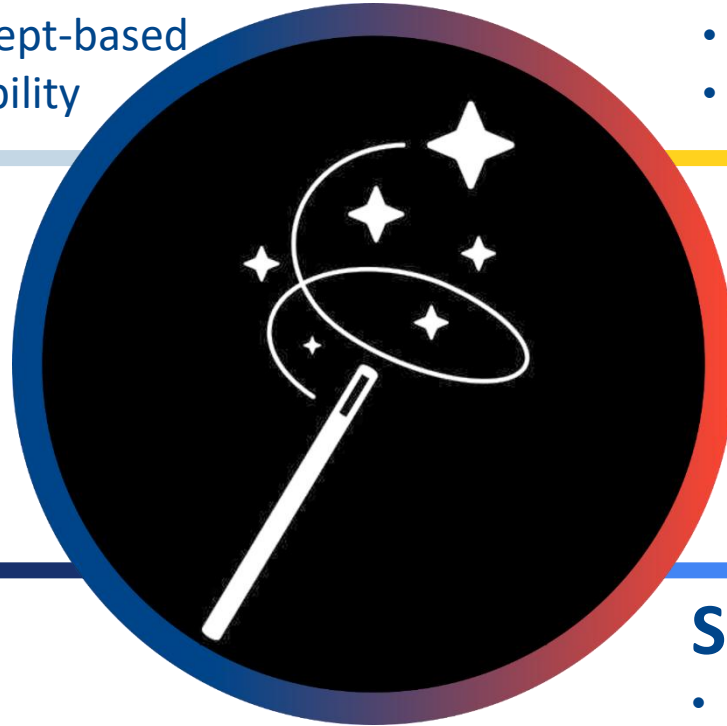
## XAI

- Attributions
- Mec Interp / Concept-based
- Post-hoc explainability



## NLP

- Any HuggingFace language model
- ForSequenceClassification
- ForCausalLM



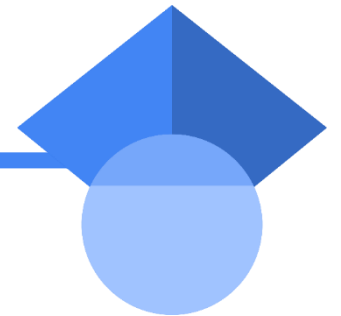
## API

- Simple & flexible
- Exhaustive documentation
- Hands-on tutorials



## SOTA

- Methods & Metrics
- Reproduce papers from the literature



# Open source

- Open Access via `pip install interpreto`
- Open feedbacks via GitHub Issues
  - Bug report
  - Feature request
  - Ask questions
- Open contributions via PR (Pull Requests)
  - Add your method
  - Propose a notebook
  - Solve an issue

# Interpreto: Differentiation

|                                           | Splitting | Learning | Interpretation | Contributions | Metrics | pip & doc |
|-------------------------------------------|-----------|----------|----------------|---------------|---------|-----------|
| <b>INTERPRETO</b>                         | ✓         | ✓        | ✓              | ✓             | ✓       | ✓         |
| <b>NNsight / NNterp / TransformerLens</b> | ✓         | ✗        | ✗              | ✗             | ✗       | ✓         |
| <b>Overcomplete</b>                       | ●         | ✓        | ●              | ✓             | ✓       | ✓         |
| <b>SAELens</b>                            | ✗         | ✓*       | ✗              | ●             | ✓       | ✓         |
| <b>Sparsify</b>                           | ✗         | ✓*       | ✗              | ●             | ✓       | ●         |
| <b>ViT-Prisma</b>                         | ●         | ✓*       | ●              | ✗             | ✓       | ✗         |
| <b>Neuronpedia</b>                        | ✓         | ✗        | ✓              | ✓             | ✓       | ●         |
| <b>Delphi</b>                             | ✗         | ✗        | ✓              | ✗             | ✓       | ✗         |
| <b>SAE-Vis</b>                            | ✗         | ✗        | ✓              | ✗             | ✗       | ✗         |
| <b>SAEBench</b>                           | ✗         | ✗        | ✓              | ✗             | ✓       | ●         |

|                   | Cls | Gen | Metrics | Simple viz | Gran |
|-------------------|-----|-----|---------|------------|------|
| <b>INTERPRETO</b> | ✓   | ✓   | ✓       | ✓          | ✓    |
| <b>Captum</b>     | ✓   | ✓   | ✓       | ✗          | ✗    |
| <b>Ferret</b>     | ✓   | ✗   | ✓       | ✗          | ✗    |
| <b>Inseq</b>      | ✗   | ✓   | ✗       | ✗          | ✗    |
| <b>SHAP</b>       | ✓   | ✓   | ✗       | ✓          | ✗    |

+ Post-hoc supervised: Probes

+ Logit Lens

# Interpreto: Future Work

- Soon (on going):
  - ViT (vision)
- Later
  - Multimodality (Text + Image) (Speech?)
  - Multi-GPU optimizations
  - Improve visualizations
  - Your contribution!

# Classification Use Case: BIOS

- Assign an occupation to a biography
- Also includes a binary gender label
- Biased Bios: Remove minority gender samples from occupations

|                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  Fannyjrd/roberta-bios-biased                                                                                                                                                                                                      |  Fannyjrd/roberta-bios                                                                                                                                                                                                               |
|  Text Classification •  0.1B • Updated about 22 hours ago •  5 |  Text Classification •  0.1B • Updated about 22 hours ago •  2 |

| Model               | Dataset     | Test Accuracy    |
|---------------------|-------------|------------------|
| RoBERTa-Bios        | Bios        | 86.89 %          |
| RoBERTa-Bios-Biased | Bios        | 85.39 % (- 1.5%) |
| RoBERTa-Bios-Biased | Bios Biased | 87.79 %          |

[De-Arteaga et al. - FAccT 2019 - Bias in Bios: A Case Study of Semantic Representation Bias in a High-Stakes Setting](#)

# List of tutorials

- Bias detection in classification
  - Application: BIOS
  - Explanations: Attributions, sample-level probes, and unsupervised concepts
- Probing POS in generation models
  - Application: POS (Parts Of Speech)
  - Explanations: Token-level probes
- Attributions for long context generation
  - RAG + reasoning

